

Design of Augmented Reality based Environment to Promote Spatial Imagination for Mathematics Education in Elementary School

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ABSTRACT

This paper investigates the use of augmented reality (AR) technology to deliver augmented lectures to support students in acquiring the curricular competency of using spatial imagination in mathematics education. As a very important stage in education to develop spatial abilities this paper focuses on elementary school children. Due to the challenges of working with children in experimental studies, this education level has received comparatively little attention in terms of support through ubiquitous technology. A theory-driven design approach was adopted in the development of an AR-based learning environment to visualize various virtual 3D cube buildings aligned with real blueprints. The designed environment was evaluated by a group of 1st and 2nd grade students in terms of system usability. The results showed that the theory-driven design was successful with a score of 86.56 on the System Usability test. In our future work, we aimed to assess the effectiveness of the proposed AR environment in terms of learning gains by performing more task-focused studies before and after experiment and comparing the results with the traditional teaching methods.

CCS CONCEPTS

• **Applied computing** → **Interactive learning environments**; • **Human-centered computing** → **Mixed / augmented reality**.

KEYWORDS

Augmented Reality, Education, HCI, Mathematics, Spatial Abilities, Usability Test

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1 INTRODUCTION

Understanding the development of spatial skills is essential to promote school preparedness and improving overall accomplishments in STEM (science, technology, engineering, and mathematics) fields [19, 20]. This is particularly important in early stages of education as positive learning outcomes associated with spatial abilities have an impact on career success in these fields [8].

Common learning materials in elementary school mathematics education to develop spatial skills are building plans and structures consisting of individual cubes. Students use these cubes to build buildings according to the presented blueprint. In this setup, the combination of verbal information, the numbers on the blueprints, and pictorial information of the cube buildings create a multimedia learning environment [12]. This information is usually presented spatially separated causing a split attention effect [3, 13]. The blocked view of the blueprints due the solid materials of the cubes (e.g., wood) in a built-on setup, or lack of integration due to separate presentation of buildings and blueprints are two main examples that cause such effect. As a suitable approach to address such challenges, augmented reality (AR) helps reducing split attention effect in multimedia learning environment by increasing spatial contiguity, and overall positively promoting the learning procedure [5–7, 18]. Although it is clear that AR based learning environments possess the potential to be beneficial, AR itself is not beneficial per se [9]. Therefore, the design and development of a new AR-based learning environment is a non trivial problem, especially for early stage education.

Even though some studies presented different AR-based approaches and applications to use for training and developing spatial skills [4, 11, 21], to best of our knowledge there are no study focusing on theory-driven development stage of application design for an AR environment to promote curricular competence of spatial imagination.

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This paper aims to develop and test a theory-driven AR-based learning environment that addresses the learning objectives required for acquisition of the curricular competence of spatial imagination during elementary school mathematics education by trying to answer the following questions:

- (1) How should an AR-based learning environment with a focus on cube buildings be designed to promote learning based on theory-driven design principles and empirical research results?
- (2) How is the developed learning environment evaluated by children in a trial with the elementary school students in terms of usability?

2 MATERIALS AND METHODS

2.1 Development of the AR-based learning environment

To successfully design an AR-based learning environment that does not hinder the learning procedure, the environment must be user-friendly and avoid causing additional cognitive load. With this goal in mind, a setup was designed to provide an AR-based learning environment to achieve the curricular competency of spatial imagination with a target group of 1st and 2nd grade school students.

2.1.1 Materials. As previously shown in various studies [10, 15], Microsoft HoloLens 2 [14] has proven to be an appropriate device when working with elementary school children, hence it was found to be useful for the intended purpose in this study. To avoid any extra difficulties caused by hand interaction in AR environment [15, 16], a Bluetooth keyboard was used to provide the interaction functionalities.

In addition, a base mat and various QR code markers were used. Each QR code marker embedded the information to generate a new building block. In this way, new buildings could be generated as needed without a need to modify the software application. Moreover, using QR codes as markers allowed optical tracking required to align the blueprints with virtual buildings. As the biggest building width used in this study was 17.5 cm (5 cubes x 3.5 cm), the location of the marker was fixed at 18 cm above the center of the mat. In this study, the fixed position of the marker was intended to avoid any unwanted movement of the marker and to help precisely align the blueprints of the buildings with virtual cube buildings. However, the information regarding the building width and the minimum distance to the virtual object could be easily embedded in the QR code to enable flexibility in positioning the marker for later use cases.

A dark grey color was chosen for the base to emphasize the visibility of the virtual object and make the marker tracking process easier as the contrast with base and marker increased. The entire setup was placed on a round, height-adjustable table. In this way, the students could walk around the table to observe the virtual object from all angles (Fig. 1).

2.1.2 Software design. During the design process of the AR software the following design goals were set to achieve theory-driven application design:

- AR should be used to simultaneously overlay the 3D cube buildings (as pictorial information) and blueprints (as verbal information) [17] to avoid split attention effects (both spatial and temporal), by addressing the multiple representation principle, and the contiguity principle.
- A creation of a mental model of 3D cube building for target group of novices that has not yet mastered spatial abilities should be aimed.
- Visual clusters on the display should be minimized [17].
- Students should be allowed to interact with the AR environment and virtual objects, however, the interactions should be controlled and limited.
- Attention should be paid to the use of inexpensive and accessible technology, and content creation should be simple.

By taking the set goals into consideration, a software application was developed for Microsoft HoloLens 2. The application was designed to demonstrate seven different building plans in a certain order. Each virtual cube building was displayed over a two-dimensional blueprint that was positioned on the table. This type of arrangement was used to facilitate the construction of a coherent mental representation and to simplify the linking between the building and the blueprint.

In addition, certain functionalities were allowed for interaction with buildings using a Bluetooth keyboard (Fig. 2). These functions allowed students to construct and deconstruct the 3D cube building by levels and towers.

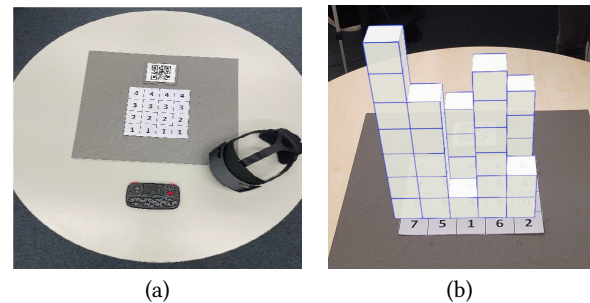


Figure 1: (a) Materials used in proposed AR-based learning environment. (b) First-person view of the app showing virtual cube building aligned with the associated real blueprint.

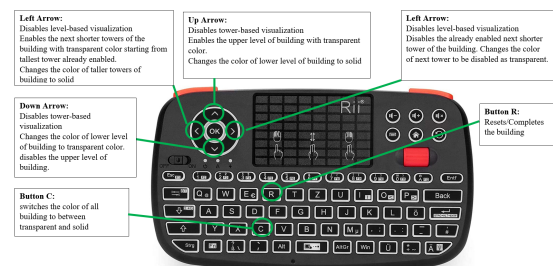


Figure 2: Functionalities associated with keyboard buttons.

2.2 Pre-test and post-test tasks

The pre-test consisted of five cube counting tasks. The post-test consisted of four type of tasks that were selected according to German mathematics curriculum for elementary school: a parallel version of the cube counting task set used in the pre-test, finding the associated building by looking at the corresponding blueprint, filling a blueprint by looking at the corresponding building, and identifying the same building from different perspectives. With five assignment for each task type, a total of 20 tasks were included in the post-test.

2.3 System stress test

A stress test was performed with a 1st grade student to test the designed software and confirm the feasibility of AR-based environment prior to the experiment. It was observed that a first-person remote view was required to fully control the experiment process and avoid a series of queries to ensure the correct visibility of the virtual content. Furthermore, an eye calibration prior to application run could help provide a better overall experience for the participant.

2.4 Experiment routine

A written consent form was filled out and signed by student's parents/guardians. A total of 16 students participated in this experiment, 10 boys and 6 girls with mean age of 7.19 ± 0.91 . One student did not complete the post experiment evaluation due to the time constraints. Hence, data of 15 participants were included in the study.

Each participant goes through the experiment according to previously defined ordered steps: introduction to the experiment and consent, pre-test, HoloLens calibration, learning phase with AR, introduction to answering questionnaires, filling out the system usability questionnaire, post-test, and collection of personal data (Fig. 3). In the pre-test session, the child was asked to solve the

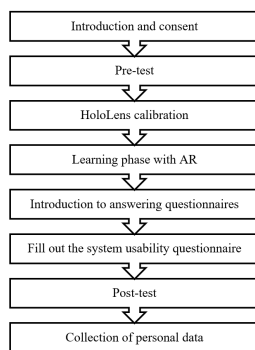


Figure 3: Experiment routine

assignment alone. However, if she/he required help in reading or understanding the question, the tutor assisted her/him through the questions. Before the learning phase, five different edge colors (dark blue, light blue, green, red, and yellow) were presented and the child was allowed to choose one color that is most visible to her/his eyes, but also does not block the numbers on the blueprints. A guided

learning approach took place afterward using various virtual cube buildings. After introduction to how to fill out questionnaires, students were asked to fill out a questionnaire on system usability. This questionnaire was designed according to Brooke et al. [2] system usability scale with an adaption for the target age group. Later, the post-test consisting of five tasks for each of the four selected task types took place. In addition, gender, age, grade, and school grades in math were asked. Finally, it was asked whether the children had already done tasks from the post-test at school.

3 RESULTS

Among all color choices, the dark blue color was preferred by far the most often for the virtual object borders.

The calculation of the system usability score was performed based on the participant's answers on a 5-point Likert scale following recommended calculation of Brooke et al. [2]. Each item's score contribution ranged between 0 and 4. For items with odd number (1,3,5,7,9) the score contribution counted as the scale position minus 1. For even items (2,4,6,8,10), the contribution counted as 5 minus the scale position. The overall system usability value then was calculated by multiplying the sum of the scores by 2.5 [2].

Using Bangor et al. [1] rating scale, the designed AR-learning environment was classified as "excellent" with a score of 86.56 on system usability scale. The overall assessment of the AR environment scored a mean of $4.81 \pm .54$, representing high agreement.

4 CONCLUSION AND DISCUSSION

The goal of the presented work was to develop and test an AR-supported learning environment on the topic of cube building and the associated competence of spatial abilities targeting elementary school children. For this purpose, attention was paid to a theory-driven design stage of AR environment.

The first research question in this study dealt with the learning-friendliness of AR environment design. The contiguity and multiple representation principle were implemented by overlaying the virtual buildings on top of real blueprint without blocking its visibility. Furthermore, a split attention effect could be counteracted by this form of presentation. However, it is worth mentioning that the lighting conditions had a great influence on the visibility of both virtual and real elements. A darkened room helped to maintain better visibility of virtual elements, however, a certain degree of brightness was necessary for both marker tracking and visibility of real objects such as blueprints. A pre-adjustment for each individual helped to find the most fitting brightness that provides the best visibility for both virtual and real objects. The edge color dark blue was selected most frequently and thus seemed to be the most visible color for the majority of the test subjects yet two participants experienced initial difficulties in recognizing the colors at all. This was remedied by changing their position in the room. In our experience, an optimal environment would be a room with dimmable light and no window or window with closed blinds. This would also eliminate the influences of the time and weather condition on the test subjects.

The second research question addressed the feasibility of the designed environment by analyzing system usability. The system usability score according to Bangor et al. [1] can be classified as

"excellent". This speaks for a successful implementation of the design principles and goals. The items regarding school context and overall assessment also achieved high values. This indicates that the subjects liked the AR-supported learning unit and would like to use it in school. Overall, the usability measures of the AR environment were very satisfactory and no adjustments are needed in this regard.

As a limitation of the designed AR-based learning unit, the designed environment lacked haptic feedback. Therefore no stimuli were received and embodied cognition could not occur. As it could be beneficial for the learning process [22], it is recommended that future designs include haptic feedback modalities. Another limitation of this study was the sample group. It should be stated that the used sample was not a complete representative sample of grades 1 and 2. More boys participated in this study. It can be also assumed that more students with a tendency of interest in mathematics agreed to participate. In the questionnaires, an attempt was made to counteract possible response biases. Nevertheless, such effects cannot be ruled out in the young target group, which is untrained and inexperienced in filling out questionnaires.

Even though the proposed system looks promising, there are still some barriers to overcome for a successful implementation of this technology in actual classrooms. The modern technologies require teachers to learn how to use these technologies in their classrooms. Hence these new technologies increase the teachers' training requirements. Moreover, the teacher's attitudes towards latest technologies are a key factor in successful implementation of supportive systems in education. Some teachers find using new technologies in education time consuming and believe students thinking abilities might be affected from using them. Another important barrier to overcome is the reliability issue. Reliability of technological systems has an important impact on education in terms of acquisition and absorption of knowledge to both teachers and students. Like all new technological advancements, perfecting a system requires time and extensive testing. Therefore, it is important to invest enough time to make sure no hardware/software failure will occur during teaching session. The cost to install latest technologies creates another important barriers to use these technologies in regular classrooms. Last but not least, due to the fact that most of the wearable devices are designed for adult users, a modification of these equipment are needed to provide better fitting for young users and prevent possible health issues.

In conclusion, the proposed theory-driven design was successful, in terms of usability. However, more researches need to be done before implementing these technologies in actual classrooms. To grasp a better understanding about the effectiveness of suggested AR environment in terms of learning gains, in our future studies we will focus on performing task-focused before/after experiment and comparing the results with traditional teaching methods.

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